

## Functional & Performance Testing

### Metro Ticket Generating System

|               |                                              |
|---------------|----------------------------------------------|
| Date          | 5 September 2026                             |
| Team ID       | SWUID2026190414                              |
| Project Name  | Metro Ticket Generating System in ServiceNow |
| Maximum Marks |                                              |

### Test Scenarios & Results

The following functional and performance test cases are prepared for the Metro Ticket Generating System. Actual Result and Pass/Fail should be updated after executing each test.

| Test Case ID | Scenario (What to test)      | Test Steps (How to test)                                                     | Expected Result                                                                                     | Pass/Fail |
|--------------|------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------|
| FT-01        | Station Selection Validation | Select valid and invalid values in Starting From and Going To fields.        | Valid metro stations are accepted and invalid/incomplete station selections are prevented.          | pass      |
| FT-02        | Journey Type Validation      | Select Single Journey and Return Journey.                                    | The selected journey type is accepted and the corresponding fare field is handled correctly.        | pass      |
| FT-03        | Passenger Number Validation  | Select the available passenger count values and test an invalid/empty value. | Valid passenger count is accepted; missing or invalid passenger count is prevented.                 | pass      |
| FT-04        | Fare Calculation Test        | Select source, destination, journey type and number of passengers.           | Fare is automatically calculated according to the selected route, journey type and passenger count. | pass      |
| FT-05        | Payment Mode Validation      | Select UPI, CARD and Others.                                                 | Each configured payment mode is displayed and can be                                                | pass      |

|       |                            |                                                                                                 |                                                                                            |      |
|-------|----------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------|
|       |                            |                                                                                                 | selected.                                                                                  |      |
| FT-06 | Ticket Submission Test     | Enter all required booking details and submit the ticket request.                               | The metro ticket request is submitted successfully with the entered journey details.       | pass |
| FT-07 | QR Ticket Generation Test  | Complete a valid booking and initiate the QR-ticket generation flow.                            | A digital QR code/ticket is generated and displayed to the passenger.                      | pass |
| FT-08 | Required Field Validation  | Leave one or more mandatory booking fields empty and attempt submission.                        | The system identifies the missing required information and prevents incomplete submission. | pass |
| PT-01 | Response Time Test         | Use a timer to check the response time when changing route, journey type and passenger count.   | Fare calculation and form response should occur without noticeable delay.                  | pass |
| PT-02 | QR API Response Test       | Trigger QR generation and observe the response from the QR-code generation API.                 | The QR API should respond successfully and the QR image should be displayed.               | pass |
| PT-03 | Multiple Booking Load Test | Execute multiple booking submissions sequentially and, where supported, from multiple sessions. | The system should continue to process bookings without errors or crashes.                  | pass |

### Testing Summary

- Functional testing covers station selection, journey type, passenger count, fare calculation, payment mode, submission, QR generation and required-field validation.
- Performance testing covers response time, QR API response and handling of multiple booking operations.